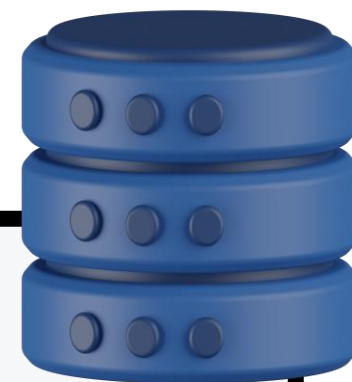




COM106: Introduction to Databases

1. Introduction & Module Overview
2. Database Basics & Security

<https://github.com/KodZone/COM106-MYSQL>

COM106 - Introduction to Databases (S1 2627): Weekly Schedule

COM106 Weekly Schedule - Lecture & Lab Delivery

Week 9 Lab is the in-class exam (Set Exercise, 90-minute class test, 40% of module mark).

Week	Lecture	Lab / Seminar
1	Module Introduction & Database Basics and Security - Course overview, aims & assessment - File-based systems & their problems - The DBMS approach: advantages & facilities - Data independence (ANSI-SPARC) & centralised data - Security threats & countermeasures - GDPR & Data Protection Act 2018	Lab 1: Getting Started with MYSQL - MySQL Server & Workbench setup / login - Creating firstdb & tables via the Workbench GUI - Primary keys & referential integrity (InnoDB) - Entering & viewing records - Exporting & importing databases (backup / restore)

Course overview, aims & assessment

<https://github.com/KodZone/COM106-MYSQL>

So What's this Lecture About?



See the **Module Handbook** on Blackboard for more information

When & Where – Timetable

Lecture

Tuesday 14:30 – 16:30 Group B

Lab Classes

Thursday 15:30 – 17:30 Group B

Who - Teaching Staff

Dr. Serif Salem - Lecture

Email: serif.salem@qa.com

Dr. Vishnu Ajith- Lab

Email: vishnu.ajith@qa.com

What will I be able to do at the end of this module?

A successful student will be able to:

- Explain the concepts underlying the design, structure, implementation, manipulation and management of relational databases in modern information systems.
- Apply appropriate models, processes and techniques in the design, implementation and use of database systems to meet an outline business scenario.
- Construct a database from a given relational schema using a commercial database management system.
- Perform basic data manipulation and information retrieval operations using SQL.

You will have:

- an understanding of the theoretical basis, structure, management and workings of database systems and provide some insight into recent developments and future trends
- a knowledge and understanding of the techniques of database design and the ability to apply these techniques to implement such designs on a relational database management system
- an understanding and practical experience of data manipulation and query in SQL
- experience in using a commercial database management system.

Week	Lecture	Lab / Seminar
1	Module Introduction & Database Basics and Security - Course overview, aims & assessment - File-based systems & their problems - The DBMS approach: advantages & facilities - Data independence (ANSI-SPARC) & centralised data - Security threats & countermeasures - GDPR & Data Protection Act 2018	Lab 1: Getting Started with MYSQL - MySQL Server & Workbench setup / login - Creating firstdb & tables via the Workbench GUI - Primary keys & referential integrity (InnoDB) - Entering & viewing records - Exporting & importing databases (backup / restore)
2	Introduction to Entity-Relationship Modelling - Database development stages - Entities & attribute types - Relationships: degree & cardinality - Participation constraints (full / partial) - ER diagram notation & worked example	Lab 2: Creating ER Diagrams with draw.io - ERD components & notations (Chen, Crow's Foot) - draw.io basics: shapes & connectors - Advanced draw.io features - Practice scenarios (Bookstore, Company)
3	ER Modelling 2 & Mapping to the Relational Model - Worked examples: attributes & data types - Connection traps: fan & chasm - Multiplicities (min, max) notation - Mapping rules: key copying & link tables - Completing the relational schema	Lab 3: ER Diagram Design from Scenarios - The ER modelling method - University Academic Management System - Library Management System - Hospital Management System - Online Course Platform
4	Normalisation - UNF & First Normal Form (1NF) - Update anomalies in 1NF - Functional dependencies & 2NF - Transitive dependencies & 3NF - Worked examples to 3NF	Lab 4: Normalisation Practice - UNF to 1NF: removing repeating groups - 1NF to 2NF: removing partial dependencies - 2NF to 3NF: removing transitive dependencies - Table decomposition exercises
5	Introduction to SQL (DDL, DML, DCL) - SQL overview & data types - CREATE TABLE & primary keys - Integrity constraints (DEFAULT, UNIQUE, CHECK) - Foreign keys & ALTER TABLE - INSERT, UPDATE & DELETE	Lab 5: Building a Database with SQL (businessdb) - Creating database & tables in SQL (MySQL data types) - Referential integrity via ALTER TABLE - ALTER TABLE variations - INSERT, UPDATE & DELETE records - Exporting / backing up businessdb (mysqldump) - First SELECT exercises (projemp)
6	SQL Queries: Single-Table & Multiple-Table - SELECT structure & <u>WHERE</u> logic - ORDER BY & DISTINCT - Aggregates, GROUP BY & HAVING - Joins: WHERE vs INNER JOIN - Nested queries & self joins - Views	Lab 6: SQL Retrieval Exercises (projemp) - The projemp schema - GROUP BY & HAVING exercises - DISTINCT & counting - Multi-table join queries - Mixed retrieval practice

Week	Lecture	Lab / Seminar
7	Further SQL: Set Operators & JOIN Queries - UNION, INTERSECT & EXCEPT - LIKE wildcards & IS NULL - JOIN types & join conditions - INNER vs OUTER joins - Finding non-matching rows	Lab 7: SQL Scripting (class_db & projemp) - Creating class_db by script - Creating tables with data types - Dropping tables in FK order - Loading & verifying projemp
8	Database Management: Transaction Management - Transactions & concurrency problems - ACID properties - Schedules & serializability - Locking & deadlock handling - Recovery: log, checkpoints & backup	Lab 8: SQL with Python in Jupyter Notebooks - Jupyter setup & packages (PyMySQL, pandas) - Connecting to projemp (MySQL connector) - Queries from Python (cursors, commit) - JOINS & result handling
9	Stored Procedures & Triggers - Stored procedures & advantages - Creating & executing procedures - Input & output parameters - AFTER triggers (Inserted / Deleted tables) - INSTEAD OF triggers & uses	Lab 9: IN-CLASS EXAM: Set Exercise (40%) - 90-minute class test held in the lab session
10	Database Management: Storage Structures - Disk access & physical design - Pages, file & disk managers - File organisation options - ISAM, B-Trees & hashing - Indexing & query performance	Lab 10: Stored Procedures & Triggers in MySQL - Setting up the proc3 database - Creating & executing procedures (DELIMITER / CALL) - Procedures with input parameters - AFTER INSERT trigger (NEW / OLD) - AFTER DELETE / UPDATE triggers (archive & audit)
11	Revision / CW2 Preparation	Lab 11: Revision / CW2 Preparation
12	CW2 Submission	Lab 12: CW2 Submission

What will we be Looking at in the Module?:-

- Database Basics
- Data Models
- Query Processing in the Relational Model
- Data Management Issues
- Running SQL in MySQL

We'll be using **MySQL** as the main Database Management Systems (DBMS) in the labs.

How Do YOU Learn the Material

Lots of material and support available:-

- Lectures & Practicals – attendance is compulsory and **engagement** is monitored
- ‘HomeWorks’ – every week, in preparation for the next week.
- Blackboard Learn: <https://learning.ulster.ac.uk>
- Library – books suitable for an Introductory Database Course
- Lots & Lots of external on-line resources available:-
 - books, blogs, videos, entire courses, simulators, etc.
 - we’ll be using some and recommending others

This module requires **200 hours total effort** (at least)

- Lectures - 24 hrs
- Lab - 24-36 hrs
- Independent Study - **140-152 hrs**. Plan to allocate between **ten to twelve hours for private study per week.**

How – Books - Library

Any introductory text on **Databases** will cover the material in this module.

- There are lots and lots available – both hardcopy and electronic

Check out the **library catalogue** available through the portal

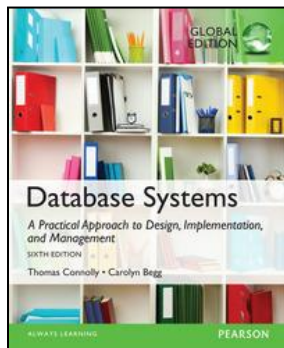
- or directly at <http://catalogue.library.ulster.ac.uk/>

And the **Subject guide for Computing** (access through the portal)

- or at <http://guides.library.ulster.ac.uk/computing>

There are a number of **recommended textbooks** for the module. It's not essential that you buy one, but it **IS ESSENTIAL** that you regularly use one!

Be aware - the slides *won't* give all the information you'll need for the module.



Database systems: a practical approach to design, implementation, and management

Connolly, Thomas M; Begg, Carolyn E

Pearson Education, 6th ed., Global ed. 2015

ISBN: 1292061189, 9781292061184

Shelfmark: **005.74/CON**

Hard copy available in the library

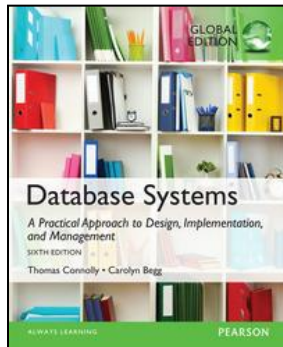
*Also available as an **ebook***

Earlier editions are also available

How – Books - Library

There are a number of **recommended text books** for the module. It's not essential that you buy one, but it **IS ESSENTIAL** that you regularly use one!

Be aware - the slides *won't* give all the information you'll need for the module.



Database systems: a practical approach to design, implementation, and management

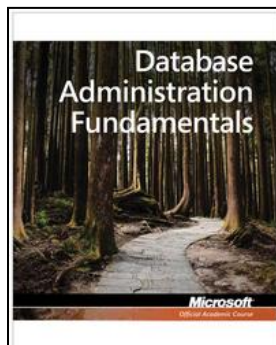
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More details are available
in the **Additional Resources Folder**



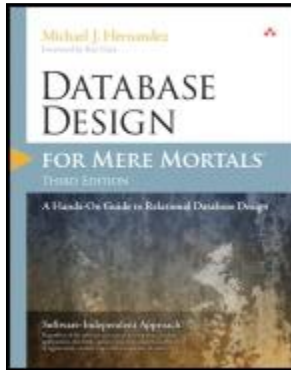
Database administration fundamentals, exam 98-364

Gambrel, Bryan; Microsoft Corporation
Wiley, 2011
ISBN: 0470889160, 9780470889169
Shelfmark: **005.74/MIC**

***These are the main text books
that will be used and referenced
throughout the module***

*An electronic copy is available on
Blackboard (Additional Resources Folder)*

How – Books - Library



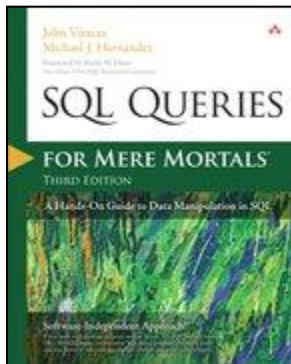
Database Design for Mere Mortals®: A Hands-On Guide to Relational Database Design

Michael J. Hernandez

Addison-Wesley Professional, 3rd ed. 2013

ISBN: 0-321-88449-3, 0-13-312228-X

*Available in the library
as an ebook*



SQL Queries for Mere Mortals®: A Hands-On Guide to Data Manipulation in SQL

John Viescas; Michael J. Hernandez

Addison-Wesley Professional, 3rd ed. 2014

ISBN: 0-321-99247-4, 0-13-382488-8

*Available in the library
as an ebook*

How – On-line Resources

Lots and Lots of external on-line resources available:-

- books, blogs, videos, entire courses, simulators, etc.
- we'll be using some and recommending others

Easily found – search Google, YouTube, etc

As we go through the course, I'll be posting the resources we recommend in the **Additional Resources Folder** in Blackboard.

If you come across something new you think would be of benefit to other students - **let me know** and I'll put it up.

How – Get Your Own Copy of MY SQL 8 (Community Free Version)

The main relational database management system used throughout this module

Download and Install instructions are available in the Lab Resources Folder on Blackboard

How is the Module Assessed?

100% Coursework - Two pieces; Set Exercise and Assignment

Assignment (60%) -

Given an outline business scenario and an unstructured file of sample data, design and implement a database capable of meeting the business need.

Submission, by Week 12 (**Date will be on the Blackboard**):

A video screencast showing a walkthrough demonstration of the code and its execution against a set of given test cases.

Set Exercise (40%) -

A 90 min class test, held during allocated labs scheduled in **Week 9** (**During Lecture of Lab time**).

Individual
Assessments.
No Group Work

How is the Module Assessed?

To Pass –

40% or more aggregated
over both
coursework components.

It is not necessary to pass each component separately.

Practical Exam (40%)	Assignment (60%)	Module Mark	Pass/Fail
60%	50%	54%	Pass
75%	25%	45%	Pass
10%	55%	43%	Pass
39%	40%	41%	Pass
40%	38%	39%	Fail
80%	10%	38%	Fail

If you miss an assessment (**for valid reasons**) – submit an **EC1**

If you **FAIL** - Resit failed components in the next year (**capped at 40%**)

File-based systems & their problems

<https://github.com/KodZone/COM106-MYSQL>

So **Why** are we learning about Databases?

First - What is 'data'?

Dictionary Definition:-

- facts and statistics collected together for reference or analysis.

Computer data is information processed or stored by a computer.

May be in a *variety of forms* from bits and bytes to text documents, images, audio clips, software programs, or other types of data.

What is a 'database'?

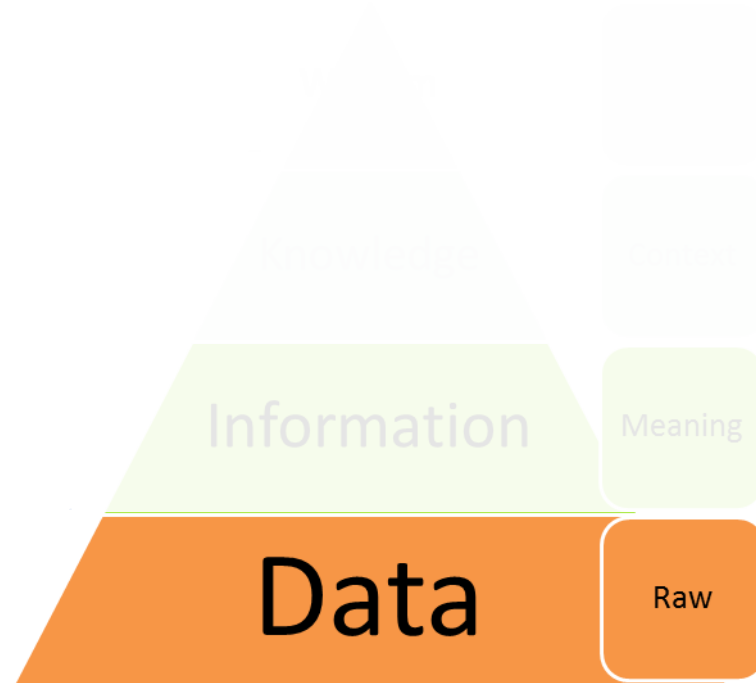
A database is an *organised* collection of data.

Information is held such that it can easily be accessed, managed, and updated.



So.... **Why** are we learning about Databases?

Data – Information – Knowledge - Wisdom



Scenario: A Weather App

Data: Raw numbers or facts

- "25°C" (temperature reading).

Information: Processed data with context

- "The current temperature is 25°C in London."

Knowledge: Applied information for understanding

- "25°C is warm for London, so wearing light clothes is appropriate."

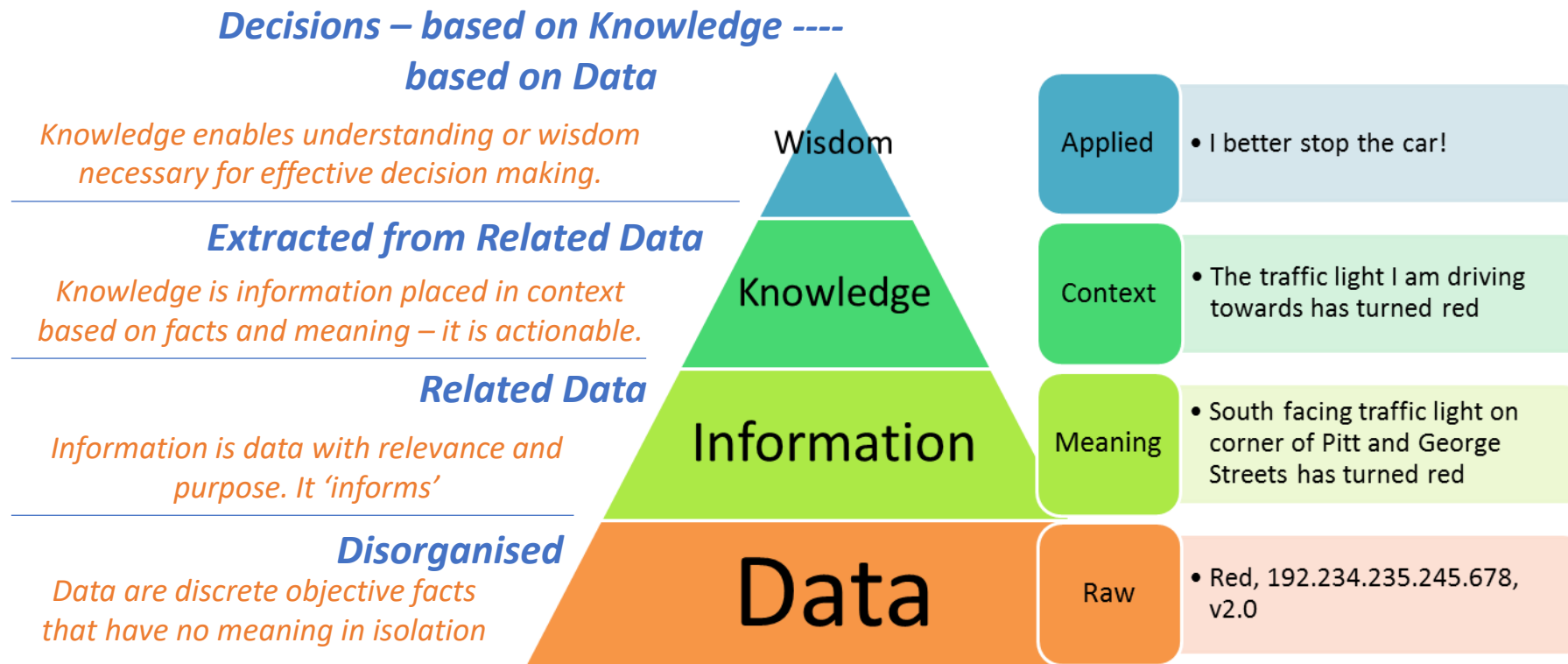
Wisdom: Use of knowledge for decision-making

- "Since it's warm, I should carry a bottle of water and wear sunscreen while heading out."

Introduction to Databases and the Relational Model

See Chapter 1, 2 & 4 of Connolly & Begg (6th ed)
(or similar chapters in other textbooks.)

Data – Information – Knowledge - Wisdom





So.... **Why** are we
learning about
Databases?

- Databases are of *fundamental importance* in a digital world.
- Necessary in any situation where:-
 - information (organised data) is needed
 - to build knowledge
 - and make decisions about future actions (wisdom)
- ***Databases are also of fundamental importance in Computing and Business, and something all Computing/Business Professionals need to know about.***



So.... **Why** are we learning about Databases?

Look up :

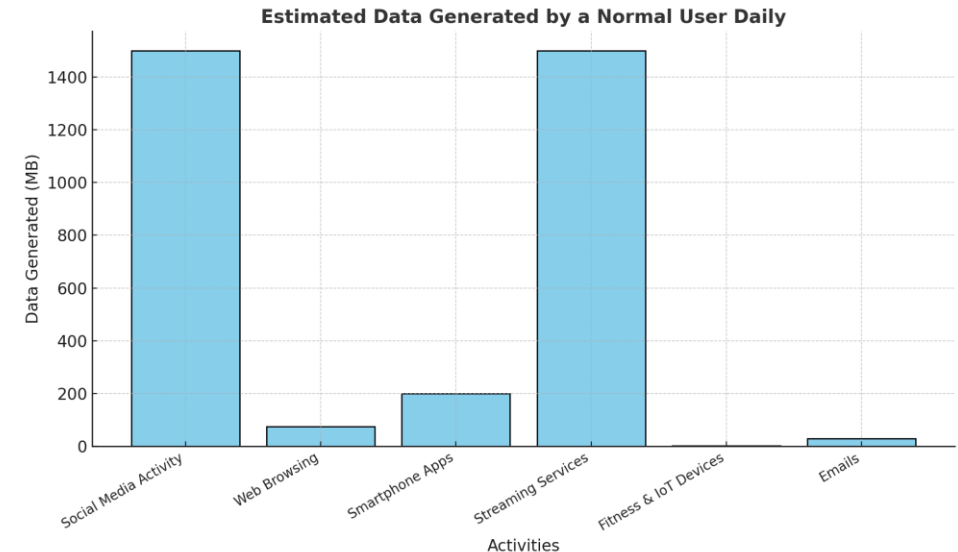
- *Data Analytics*
- *Big Data*
- *Data Mining*
- *Zettabyte*

There's an unimaginably large amount of data in the digital universe :

- estimated at **16 Zettabytes** (10^{21} Bytes) in 2017, 90% of which was created in the last two years.
 - *For more info, and some truly staggering stats, google 'amount of data in the world 2023'*
- Increasing need to analyse data in **near real-time** to derive **value** from it

Scenario: How Data is Big

- On average, a single individual generates 1.7 MB of data per second.
- Including:
 - Social Media Activity
 - Web Browsing
 - Smartphone Usage
 - Streaming Services
 - Fitness and IoT Devices
 - Emails



Total Estimated Daily Data

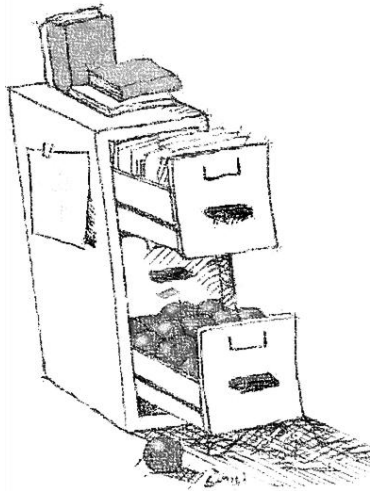
- A regular user can generate **500 MB to 2 GB** of data per day, depending on their usage patterns.
- This data is collected by companies to analyse user behaviour, preferences, and improve personalised services.

Introduction to Databases & the Relational Model

So, whilst **DATA** is a **disorganized** collection of facts, numbers, audio clips, images, etc

a **DATABASE** is:

- a shared **collection of logically related data**
- stored in an **organised structure**
- which provides the ability to **interact** with data
- and to extract salient **information and knowledge**.



The Evolution of Databases – Manual Filing Systems

- An organised collection of documents in separate files – logically related
- Held in locked filing cabinets, possibly over different premises
- Data can be manipulated – accessed, deleted, updated, etc
- And can be searched - first entry / indexing system
- Can work well if there are a small number of entries or.... a large number of entries, but used to store and retrieve only.
- **BUT** can be difficulties for anything more complex
- Data is distributed (**decentralised**) and may contain **duplicates**
- May be difficult to extract **information and knowledge**.
e.g., *Total Annual Salary Bill across all Departments*

Introduction to Databases & the Relational Model

The Evolution of Databases – File-Based Approaches

First attempt at computerisation – modelled on the manual filing system

Initially, an **application-centred** approach, where an application program worked on associated data files – e.g., a **payroll system**.

Later, files could be shared among applications to reduce **data duplication**.

PROBLEMS:

- No data **centralisation** (leading to data isolation and separation)
- Poor file design can lead to **data duplication** (or data redundancy).
- Updating multiple record copies is inefficient and may lead to loss of **data integrity**
- **Data dependence** – some data in the system may be dependent on others and should be updated together
- Application programs need a knowledge of file storage structures (queries are not independent of the data files or storage structures)
- Incompatible file formats
- Security (file level vs. data item level)
- Data organisation and structure is limited so limited queries are possible

Introduction to Databases & the Relational Model

Explanation of Terms – Data Integrity

Accurate data is vital in all information systems

Data integrity is concerned with:

Consistency (is duplicate data identical?)

Correctness (is the stored data accurate?)

Maintaining data integrity involves:

ensuring data is **designed correctly** (data model)

ensuring data is **entered correctly** (validation)

ensuring data is free from **corruption**, **modification** or **unauthorised disclosure** (programming errors or hacks)

ensuring data correctness is **maintained** (backup, recovery, transaction integrity maintenance)

Using
Integrity Constraints



Explanation of Terms – Data Dependency

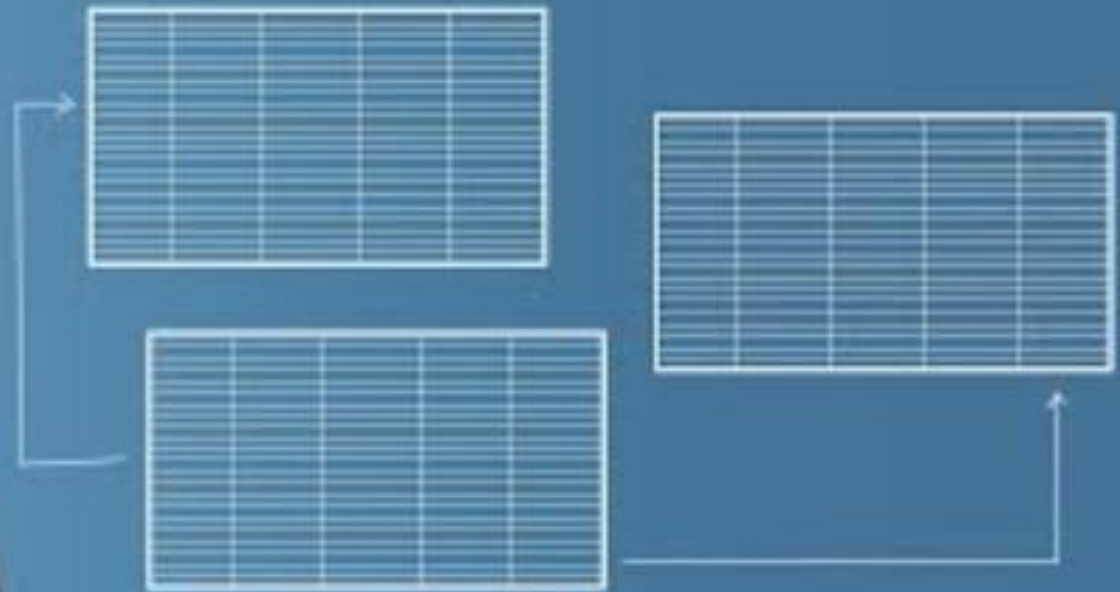
A dependency occurs when information stored in a database table (or in a file) **uniquely determines** other information stored in the same table (or file).

i.e., knowing the value of one piece of data is enough to tell you the value of another in the same table (or file).

The DBMS approach: advantages & facilities

<https://github.com/KodZone/COM106-MYSQL>

DATABASE OVERVIEW



Database Basics (continued) & Data Models

Some Examples

Given the data table below, identify any **data dependencies** by determining all of the statements that are **TRUE**

- A – Knowing **EName**, uniquely identifies **Manager**
- B – Knowing **Manager**, uniquely identifies **Dept#**
- C – Knowing **Salary**, uniquely identifies **DName**
- ✓ D – Knowing **DName**, uniquely identifies **Manager**
- E – Knowing **EName**, uniquely identifies **Dept#**
- F – None of the above

Is the reverse True?

Does knowing **Manager** uniquely identify **DName** ?

No

Does the table contain any more data dependencies?

Yes:-

Dept# gives **DName** gives **Manager**

Notice the **data duplication** in the table

Poor database design

Emp#	EName	Salary	Dept#	DName	Manager
1	Smith	15000	A	Sales	Kelly
2	Jones	21000	B	Admin	Whyte
3	Kane	18000	D	Finance	Coyle
4	Dwyer	20000	B	Admin	Whyte
5	Cook	16000	C	Personnel	Woods
6	Healy	15000	A	Sales	Kelly
7	Conroy	19000	B	Admin	Whyte
8	Price	22000	D	Finance	Coyle
9	Murphy	23000	C	Personnel	Woods
10	Jones	16000	A	Sales	Kelly

Database Basics (continued) & Data Models

An example of some of the problems of the file-based approach:

Consider a University, where each department maintains and processes its own data.

Student Accommodation Records

FName	SName	Acc Type	Address	Grades
Joe	Blogs	Halls	Room D1	CCD
Mary	Smith	Flat	10	AAB
Jane	Jones	Halls	Room A3	A*CB
David	Banda	Flat	23	210

Accommodation Staff Records

Name	Position
Jennifer	Inspector
Pat	Inspector

Student Registration Records

FName	SName	Studt No	Address	Grades
Joe	Blogs	B00123	Room D1	CCD
Mary	Smith	B00456	Room A1	AAB
Jane	Jones	B00789	Room A3	A*CB

Payroll Records

Name	Position
Jennifer	Inspector
Pat	Cleaner

Database Basics (continued) & Data Models

Student Accommodation Records

FName	SName	Acc Type	Address	Grades
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Payroll Records

Name	Position
Jennifer	Inspector
Pat	Cleaner

Data Duplication

Data Consistency

Data Correctness

Security

No Data Centralisation
Possible incompatible file formats

Data Consistency

Database Basics (continued) & Data Models

The Evolution of Databases – Database Approaches

The database approach overcomes many of the problems with the file-based approach

(From earlier) a **database** is a shared **collection** of **logically related data** stored in an **organised structure** which provides the ability to **interact** with data and to extract salient **information and knowledge**.

A database provides a **data-centred** approach, designed to meet the needs of an organisation, where data can be **shared** among different applications.

A **database management system (DBMS)** is
a software system for managing a database(s)
which allows the creation, manipulation (queries) and maintenance of data
and provides controlled access to the data (security, integrity, recovery etc).

Examples of common DBMS include:

Microsoft Access - Single user DBMS, supports limited database size

Microsoft SQL Server - Multiuser, supports larger databases

MySQL – Free, supports larger databases, often used as database server in web-based applications

Oracle - Large scale DBMS in multiuser environments, expensive, many supporting facilities

Database Basics (continued) & Data Models

Database Advantages (over File Handling Systems)

1. **Data Independence** - application programs isolated from data
2. **Centralised Data** - security and integrity maintained
3. **Access Flexibility** - query language and storage mechanisms
4. **Data Model** - reflects real world (network, hierarchical, relational etc.)

We'll look at each of these in more detail

Other DBMS Advantages

Enforcement of **Standards**

Data Sharing (different applications built on same data)

Economy of Scale (all applications built on common source data)

Increased Productivity (DBMS provides **many standard functions**)

.... and Disadvantages

Complexity (complex system requires level of understanding)

Cost (in large organisation with large number of records, cost of DBMS plus conversion and maintenance costs are high)

Higher **impact of failure** (*downside of centralisation*)

Database Basics (continued) & Data Models

DBMS Facilities

- Application Programming Language – e.g. Visual Basic in MS Access
- Report Generator (data analysis for management)
- Data Dictionary (stores data definitions etc.)
- independent of the stored data (supporting data independence)
- Query Language - e.g. **SQL** in relational database DBMSs
- Screen/Form Editor - allows creation of data entry screens (**views**)

Functions provided by DBMS

- Data storage, retrieval & update
- System Catalog/Data Dictionary - definition of all data objects
- **Transaction** support - ensures correct update completion
 - transaction **concurrency control** for multiuser access
 - transaction **recovery** from failure of database
- User authorisation
- Provide access across a network
- Provides **data integrity** and promote **data independence**
- And a range of utilities (e.g. **Indexing**, Performance monitoring, file import/export)

Data independence (ANSI-SPARC) & centralised data

<https://github.com/KodZone/COM106-MYSQL>

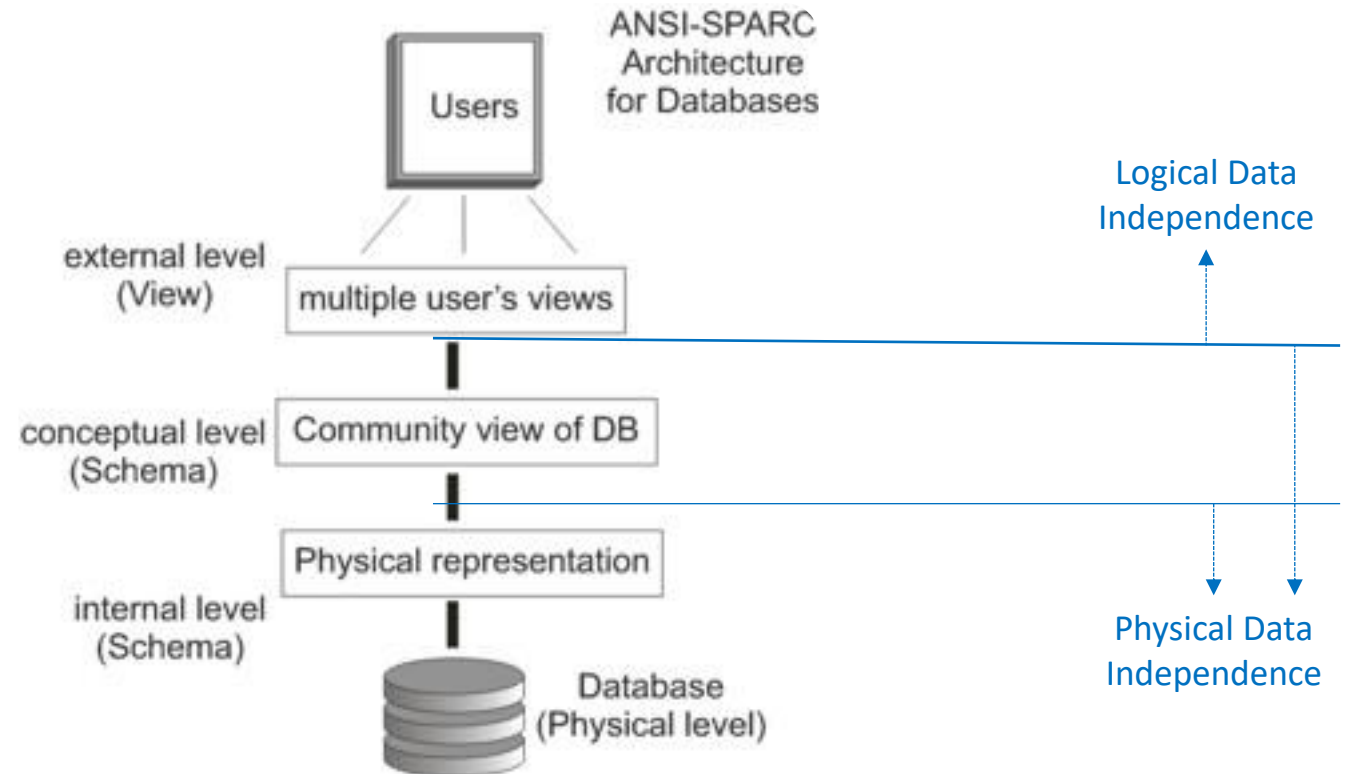
Database Basics (continued) & Data Models

Database Advantages - 1. Data Independence

Applications **separated** from data using a **3-layered model (ANSI SPARC)** of a database:

- **External Layer (Views)**: programs, queries, reports
- **Conceptual Layer (Database Schema)**: single representation of logically stored data (e.g. **set of linked relational tables**)
- **Internal Layer (Physical)**: definition of how data is stored and access paths to that data

Each layer is defined by a **schema** (set of definitions)



ANSI-SPARC stands for:
**American National Standards Institute –
Standards Planning and Requirements Committee**

Three levels of **abstraction**

Database Basics (continued) & Data Models

Data Independence Explanation:

- The description of the database structure is the **database schema** – like a blueprint of how the database is constructed
- Changes to the applications (queries/programs etc) do not necessarily require changes to the tables or stored records (**logical independence**)
- Changes to the record storage or table design do not necessarily require changes to the applications (**physical independence**)

Database Advantages - 2. Centralised Data

DBMS is managed by a **Database Administrator**

Data is stored centrally, thereby allowing/requiring:

Security to be applied to users and database objects as appropriate (passwords, privileges, views)

Integrity of the data to be maintained through:

Integrity Constraints (e.g., input validation, referential integrity, etc)

transaction management

- concurrency control
- recovery (backup + automated recovery)

*We'll study these topics
later in the module*

Database Basics (continued) & Data Models

Brief Aside – Transaction Processing

A **transaction** (key concept) is one or more **atomic sequential operations** that make up a single task

Operations - one of four categories (**CRUD**)

- (1) **C**reate - (2) **R**ead (Query) - (3) **U**update - (4) **D**delete

An Example of a Transaction – ATM Withdrawal

Checks the Account Balance (**R**ead)

Account Table

Account#	Balance
123456789	£300



Reduces balance by withdrawal (**U**update) *Update Account – Set Balance - £20*

Logs the transaction (**C**reate)

Log Table

Transaction#	Account#	Action	Date
TR_89	123456789	-£20	28/11/15

Database Basics (continued) & Data Models

Database Advantages - 3. Access Flexibility

Powerful **query language** to allow ad hoc (immediate) querying of the database

- Query By Example - Easy to create visual queries
- **SQL** - powerful query language (starting in upcoming weeks)

We'll study these topics later in the module

Storage structures and access paths

- e.g. Indexed Sequential, Hashing, B-Trees

Physical storage reorganisation (for large databases)

- in response to query performance (query optimisation)
- might involve a change of storage structure

We won't be studying these topics

Database Advantages - 4. Data Model

Data Model – How the database is organised to reflect the **real world**

Important real-world objects (**entities**), and the associations and **relationships** between them, are modelled.

e.g. **Employee** **works_in** **Department**
Student **enrolls_in** **Module**

We'll study ER Modelling later in the module

Database Basics (continued) & Data Models

Different data models used:

Hierarchical

Network

Relational – uses the **Relational Model** to give flexible navigation through data relationships between tables

Object oriented

Look at these data models in your own independent study

The vast majority of databases in use today are Relational Databases

NoSQL are a new class of database system that is growing in popularity **for particular tasks**

For NoSQL --- read --- **‘Not only SQL’**

Lots of different variations, and lots of different products (e.g., MongoDB, CouchDB, Neo4J, etc)
Document Store; Key Value Store; Graph Databases

Designed to address problems which relational database are not good at tackling

Need for a **flexible schema**

Dealing with **enormous** amounts of **unstructured/semi-structured data** (Big data)

Where the ability to **scale** is more important than maintaining **data consistency**.

*We focus on the **Relational Model** in this module.
We won't look at **NoSQL** in much detail.*

Security threats & countermeasures

<https://github.com/KodZone/COM106-MYSQL>

Database Management - Security

Some Background – Why is Security Important?:

*See Chapter 20 & 21 of Connolly & Beggs (6th ed)
(or similar chapters in other textbooks.)
and Lesson 5.1 in Database Administration Fundamentals*

- LinkedIn -- June 2021 Impact: 700 million users
 - Professional networking giant LinkedIn saw data associated with 700 million of its users posted on a dark web forum in June 2021, impacting more than 90% of its user base.
- Data is a **valuable resource** that must be strictly controlled and managed, as with any corporate resource.
- Part or all of the corporate data may have **strategic importance** and therefore needs to be kept secure and confidential.
- **Countermeasures are mechanisms** that protect the database against intentional or accidental threats.
- Depending on the nature of the data, there may be **legal, ethical** and/or **regulatory** requirements to keep data secure.

<https://www.csoononline.com/article/2130877/the-biggest-data-breaches-of-the-21st-century.html>

Database Security - Threat

*The protection of the database against **intentional** or **unintentional** threats using **computer-based** or **noncomputer-based** controls*

*See Chapter 20 & 21 of Connolly & Beggs (6th ed)
(or similar chapters in other textbooks.)
and Lesson 5.1 in Database Administration Fundamentals*

Measures should be in place to guard against threat:

- Theft and fraud
- Loss of confidentiality (organisational secrecy)
- Loss of privacy (data about individuals)
- Loss of integrity (correctness of data)
- Loss of availability (available for use)

In cybersecurity, information security should comply to the three core principles known as **CIA**.

1. Confidentiality  - Data is accessible only to authorized individuals.

- **Countermeasures:** Encryption, access controls, multi-factor authentication (MFA).

2. Integrity  - Maintains the accuracy and consistency of data.

- **Countermeasures :** Hashing, checksums, digital signatures.

3. Availability  - Ensures that **data and systems** are accessible when needed.

- **Countermeasures :** Redundancy, backups, RAID, disaster recovery plans.

Database Management - Security

Any situation or event, whether intentional or accidental, that may adversely affect a system and consequently the organisation, must be **identified** and steps taken to **reduce** or **remove** the impact.

- Organisations will have a **Business Continuity Plan** and a **Disaster Recovery Plan** in place.
- Given their fundamental importance, Databases are a crucial aspect of such plans.

Types of threat:

Hardware

- Fire/Flood/Disaster
- Power Supply Failure/Surge
- Physical Security Failure (Loss/Damage)

See also Table 20.1 Connolly & Beggs for further examples.

Software

- Virus/ Worm/ Trojan /Malware etc

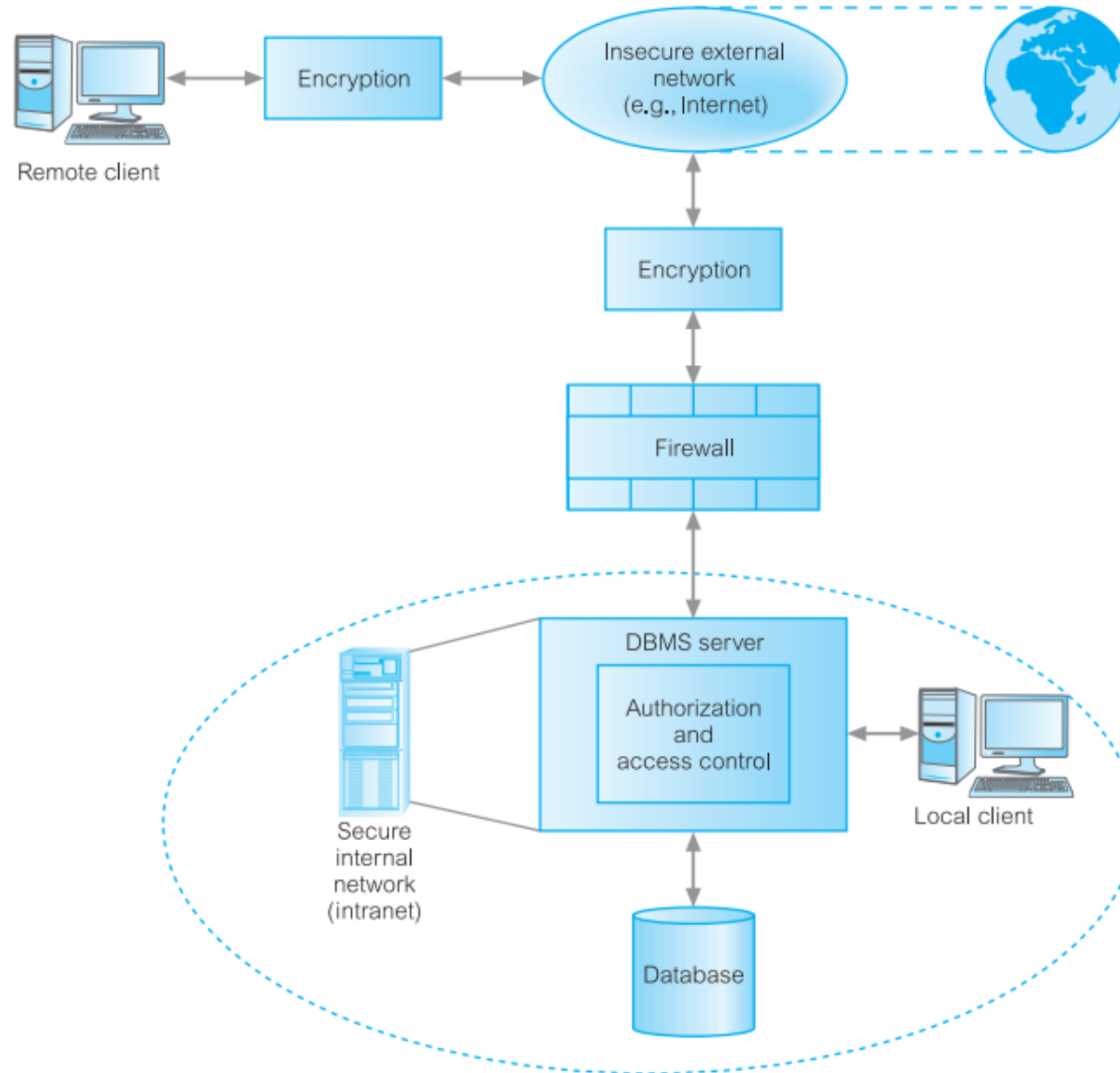
Communication

- Message interception

Users

- Unauthorised Access (Industrial Espionage/Data Protection)

Typical Multi-user Computer Environment



Cybersecurity – Computer-Based Controls: Countermeasure

Countermeasures are the actions to ...

- Protect sensitive information from unauthorized access, data breaches, and other cyber threats.
- Some examples of cybersecurity measures and best practices **for databases**:
 1. Authorization
 2. Access Control
 3. Views
 4. Backup and Recovery
 5. Integrity
 6. Encryption
 7. RAID

Countermeasures – 1. Authorization

Authentication vs. Authorization

1. Authentication → Verifies **who** the user is (**identity verification**).
2. Authorization → Determines **what** the user is allowed to do (**Level of permissions**).

Authorization is the **granting of permissions** that enable a user (subject) to **legitimately access** a system or its objects.

It is a **mechanism that controls access** to resources, ensuring users can only interact with data, files, or functionalities for which they have explicit permission.

Examples of Authentication Methods

- Something You Know → Passwords, PINs
- Something You Have → Keycards, Security Tokens
- Something You Are → Biometric (Fingerprint, Face ID, Retina Scan)

Authentication



Confirms users are who they say they are.

Authorization



Gives users permission to access a resource.

Countermeasures – 2. Access Control

- **Access control** is a security mechanism that manages user permissions by **granting or revoking privileges** to ensure proper data access.
- A **privilege** allows a user to **create, read, write, or update** data, enabling them to perform necessary job-related tasks.



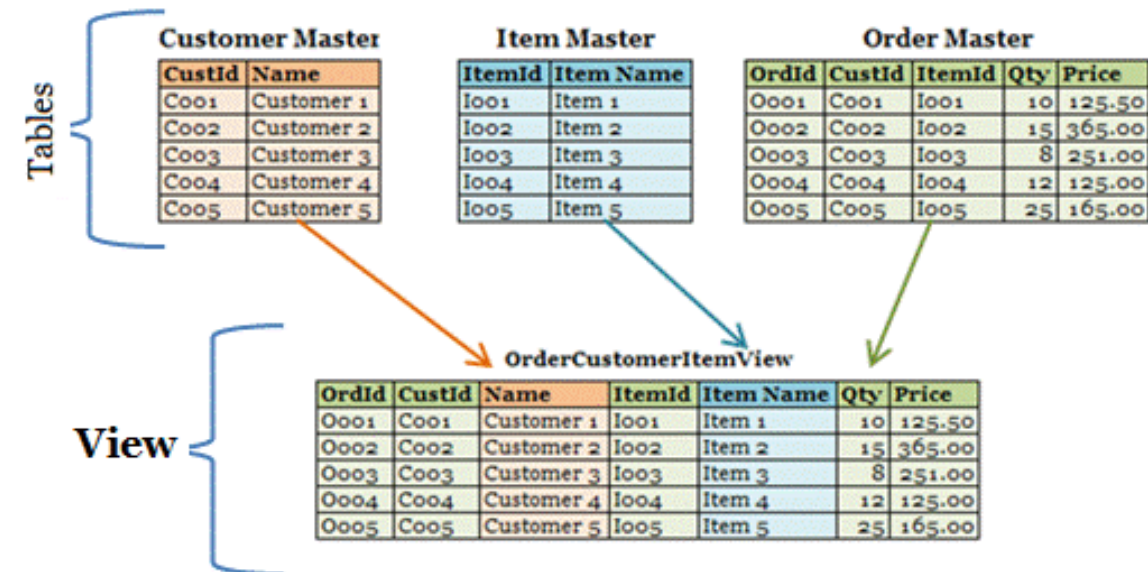
- **Discretionary Access Control (DAC)** is an access control model where the **owner of an object** (e.g., a table or database) has the **authority to grant or revoke access** to other users at their discretion.
- Most **Database Management Systems (DBMS)** implement **DAC** using the **GRANT** and **REVOKE** commands, allowing **fine-grained control over database access**.

Countermeasures – 3. Views

- A **view** is a virtual table created from one or more base tables.
- **Characteristics:**
 - Does **not store data** physically.
 - **Dynamically generated** when accessed.
 - Simplifies complex queries.
 - Enhances **security** by restricting access to specific data.

Benefits of Views:

- ✓ **Data Security** – Restrict access to sensitive columns.
- ✓ **Simplicity** – Predefine complex queries.
- ✓ **Consistency** – Provides a stable interface even if the underlying schema changes.



Countermeasures – 4. Database Backup & Journaling

Backup

- Periodically creates a copy of the **database** and **log files** for offline storage.
- Protects data from **accidental loss, corruption, or system failures**.
- **Types:**
 - **Full Backup** – Copies the entire database.
 - **Incremental Backup** – Saves only changes since the last backup.
 - **Differential Backup** – Saves changes since the last full backup.

Journaling

- Maintains a log of all database changes.
- Enables **data recovery** in case of failures.
- Records **INSERT, UPDATE, DELETE** operations for rollback or redo.

What is Data Integrity?

Ensures that **data remains accurate, consistent, and reliable** over its lifecycle, preventing invalid or incorrect information.

Why is it Important?

- Prevents **data corruption and inconsistencies**
- Ensures **accurate query results**
- Supports **reliable decision-making**

Data integrity in DB can be achieved through DB design, such as:

- **Choose the right Keys**; primary key and foreign keys.
- **Remove data duplication** by use normalisation so each fact is stored only once.
- **Group related attributes correctly** by use functional dependency so attributes that depend on each other stay in the same table (); unrelated attributes are split out.
- **Apply constraints in the schema** – define NOT NULL, UNIQUE, CHECK and DEFAULT rules at design time

What is Encryption?

Encryption is the **process of encoding data** using a special algorithm, making it **unreadable** without a decryption key.

Why is Encryption Important?

- **Protects sensitive data** from unauthorized access
- Ensures **data confidentiality** in storage and transmission
- Helps meet **security compliance regulations**

Common Encryption Methods:

- AES (Advanced Encryption Standard)
- RSA (Rivest-Shamir-Adleman)
- DES (Data Encryption Standard)

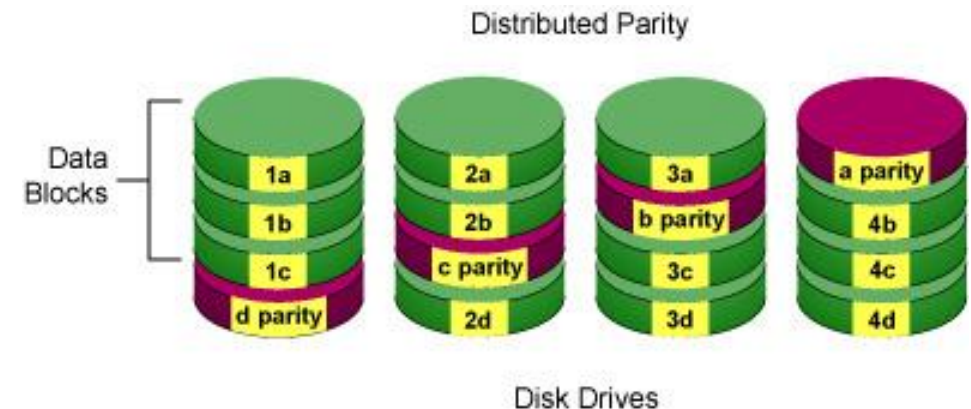
Countermeasures – 7. RAID Technology

RAID (**Redundant Array of Independent Disks**) is a **fault-tolerant** storage solution that ensures **data availability** even in case of hardware failure.

- **Ensures system reliability** – Continues operation even if a disk fails
- **Provides data redundancy** – Prevents data loss with multiple copies
- **Enhances performance** – Distributes data across multiple disks for faster access

RAID Levels & Their Benefits:

- **RAID 0** – Striping (Improves speed, no redundancy)
- **RAID 1** – Mirroring (Complete data backup)
- **RAID 5** – Striping + Parity (Balanced speed & fault tolerance)
- **RAID 10** – Mirroring + Striping (High performance & redundancy)



Recommendation: Use RAID to integrate **redundant components** that seamlessly take over in case of failure, ensuring uninterrupted DBMS operation.

GDPR & Data Protection Act 2018

<https://github.com/KodZone/COM106-MYSQL>

Database Security – Legal Requirements

IT professionals must be aware of **the laws and regulations** that affect how data may be **collected**, **processed**, **stored** and **distributed**.

Each country, jurisdiction and/or organisation will have their own requirements.

In the UK, the legal requirements include compliance with:

- **EU General Data Protection Regulation (GDPR)**

- Before the GDPR, the EU Directive on Data Protection (1995) as implemented under its own national laws.
- Adopted by the EU: 27 April 2016
- Came into force: 24 May 2016
- Became fully applicable/enforceable: 25 May 2018
- Dealing with the protection of individuals with regard to the **processing** of **personal data** and on the gathering, storage and movement of such data.
- Based on seven principles, dealing with:

Data which allows **anyone** to link info to a specific person.

Any manual or automatic operation on personal data

1) *Lawfulness, fairness and transparency,*

2) *Purpose limitation,*

3) *Data minimisation,*

4) *Accuracy,*

5) *Storage limitation,*

6) *Integrity and confidentiality,*

7) *Accountability*

Data Protection Act 2018 (DBA 2018) is the UK Data Protection Act

- Data Protection Act 1984 — the UK's first major data protection legislation.
- Data Protection Act 1998 — replaced the 1984 Act.
- Data Protection Act 2018 — replaced the 1998 Act and works alongside the GDPR.
- Together with GDPR, forms part of the data protection regime in the UK

Freedom of Information Act:

- Public-sector organisations, such as government departments and councils, must be able to find and provide information when legally requested by the public.

GDPR and the Data Protection Act 2018:

- Give individuals greater **control over their personal data**.
- Require organisations to **protect and responsibly process personal data**.
- Include rights such as the **right to be forgotten**.
- Apply strict rules for **sharing/transferring personal data**.
- Breaches can result in **significant financial penalties**.

GDPR – The Seven Key Principles

Article 5(1) and Article 5(2) of the GDPR requires that personal data shall be:

1. **Lawfulness, Fairness, and Transparency** – Personal data must be processed legally, fairly, and transparently, ensuring individuals understand how their data is used.
2. **Purpose Limitation** – Data should only be collected for specified, explicit, and legitimate purposes and not used beyond those purposes.
3. **Data Minimization** – Only the necessary personal data should be collected and processed to achieve the intended purpose.
4. **Accuracy** – Personal data must be accurate, kept up to date, and corrected without delay if found incorrect.
5. **Storage Limitation** – Data should be kept only for as long as necessary and securely deleted when no longer needed.
6. **Integrity and Confidentiality (Security)** – Organizations must protect personal data from unauthorized access, loss, or damage using appropriate security measures.
7. **Accountability** – Data controllers are responsible for compliance with GDPR and must demonstrate adherence to these principles.

GDPR & DPA (2018) – Database Implications

These Principles have fundamental implications for database designers and administrators.

The principles of accuracy and storage limitation have implications for:

- Data Input verification and validation
- Transaction Management
- Database backup/recovery
- Archiving policy – what should be kept, and for how long?

The Principle of Integrity and Confidentiality means that:

- Database security should be **designed** and **organised** to fit the nature of the personal data held and the harm that may result from a security breach.
- There should be **clarity on who is responsible** in the organisation for ensuring information security.
- Appropriate physical and technical security should be in place (**eg, authorisation and access controls, encryption, etc**), backed up by robust **policies and procedures** and reliable, **well-trained staff**.
- Processes should be in place to respond to any breach of security swiftly and effectively.

Some Definitions:

Data Controller - determines the purposes and means of processing personal data.

Data Processor - responsible for processing personal data on behalf of a controller.

Data Subject - a person whose personal data is processed by a controller or processor

Personal Data - any information related to a Data Subject, that can be used to directly or **indirectly** identify the person

Sensitive Personal Data - special categories of personal data that are subject to **additional protections**, including: racial or ethnic origin, political opinions, religious or philosophical beliefs, trade-union membership; data concerning health or sex life and sexual orientation; genetic data or biometric data.

Personal Data Breach - a **breach of security** leading to the accidental or unlawful access to, destruction, misuse, etc. of personal data. GDPR **requires that data breaches are reported** to the Information Commissioner; previously, the DPA 1998 did not.

Data Protection Officer - employed by an organisation and responsible for overseeing data protection strategy and implementation to ensure compliance with GDPR requirements.
Legal requirement in some organisations

Information Commissioner's Office (ICO) - the independent regulatory office (national data protection authority) dealing with the DPA 2018 and GDPR.



**THE
END**